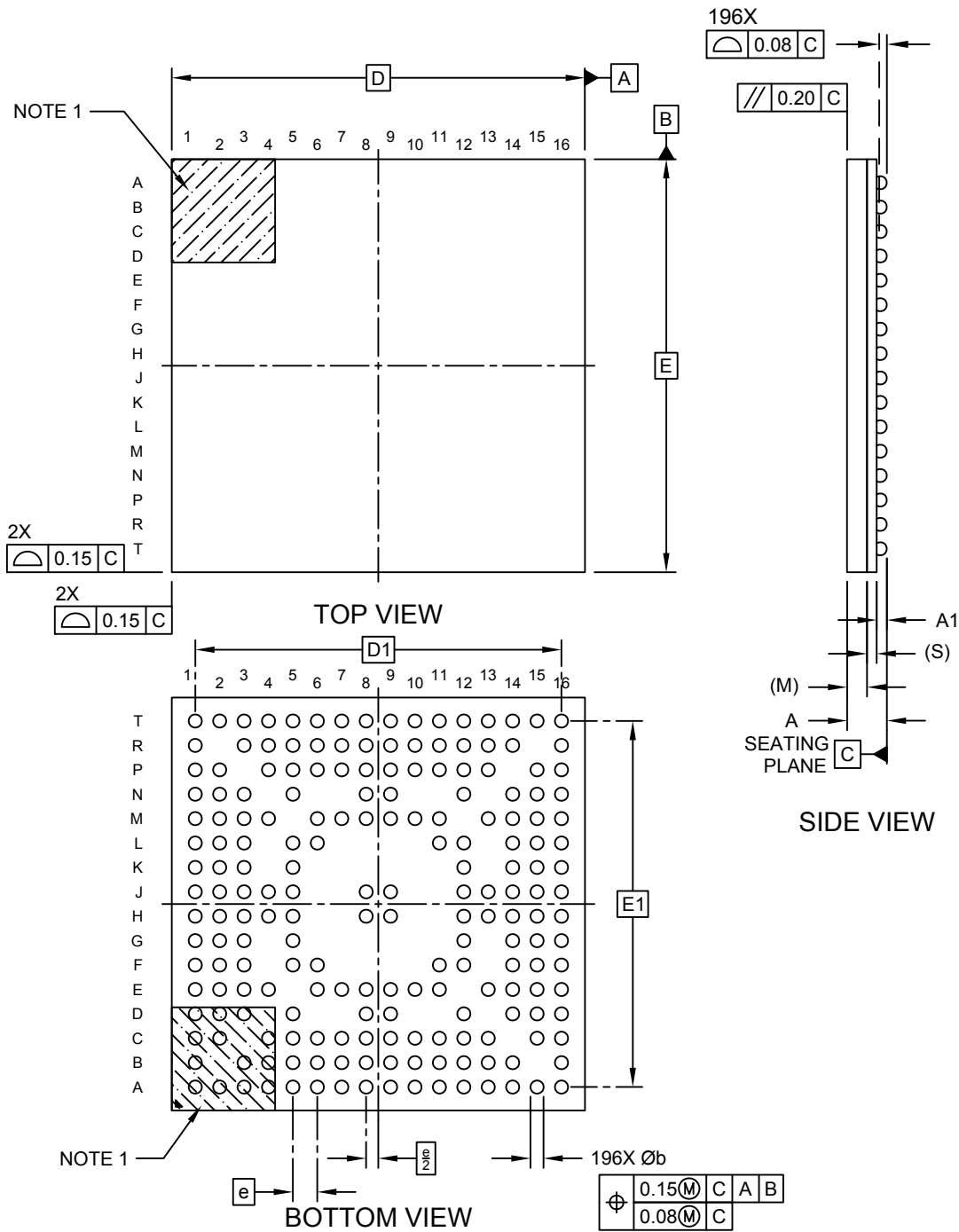


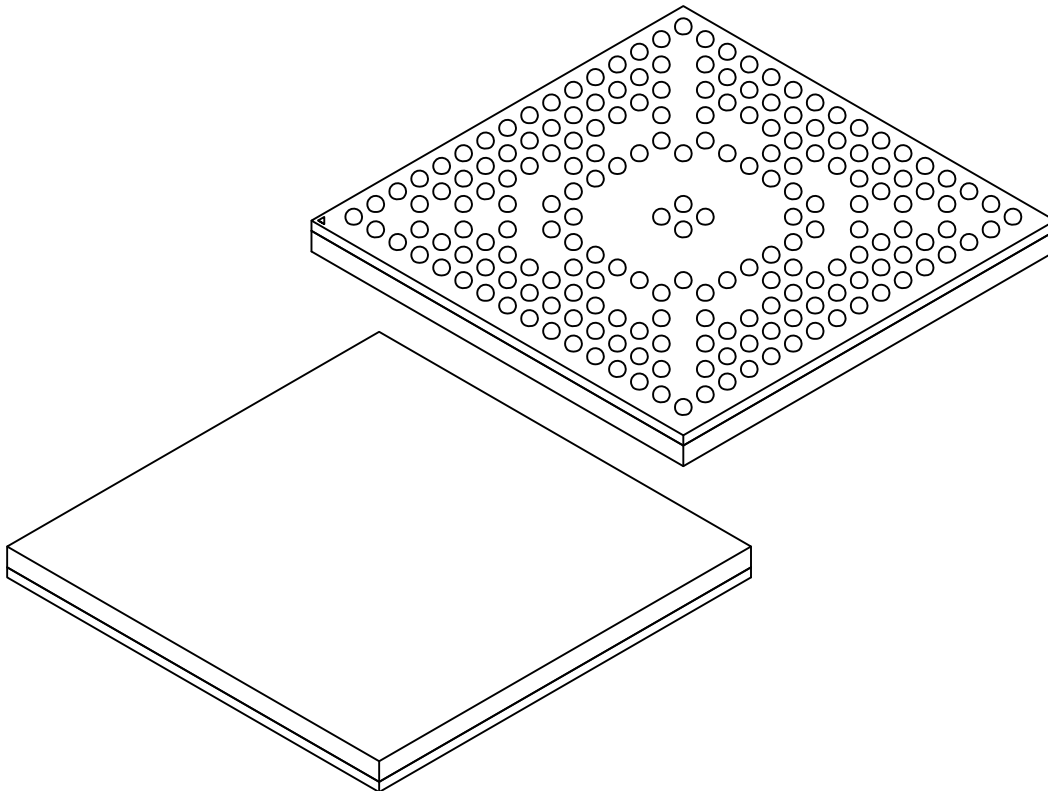
196-Lead Thin Fine Pitch Ball Grid Array (4GB) - 11x11x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



196-Lead Thin Fine Pitch Ball Grid Array (4GB) - 11x11x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



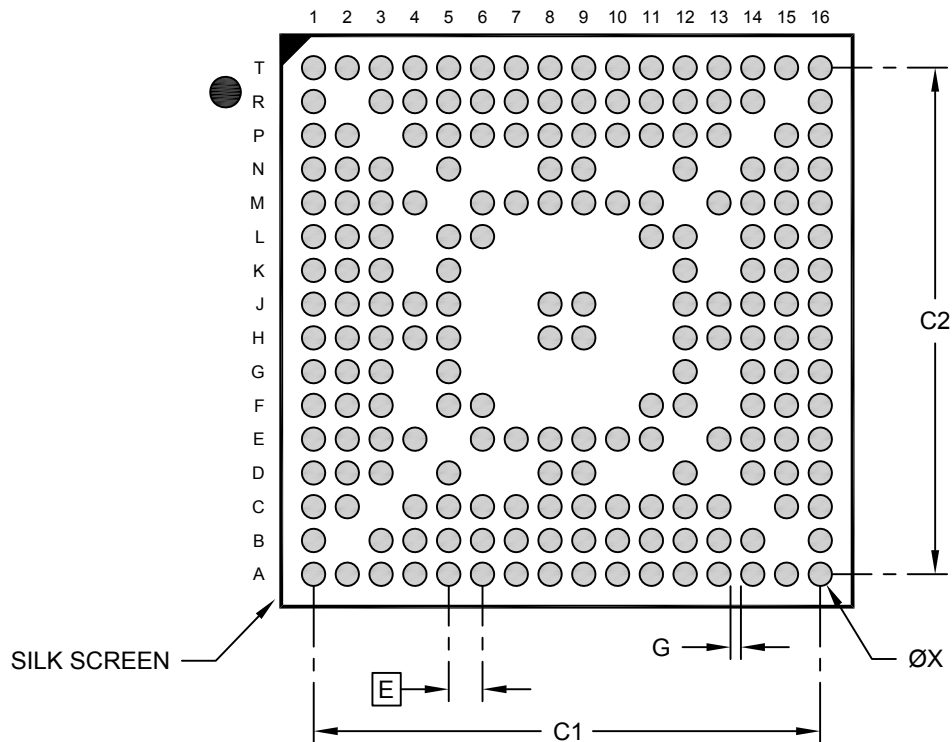
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	196		
Pitch	e	0.65 BSC		
Overall Height	A	-	-	1.20
Standoff	A1	0.22	-	0.32
Substraight Thickness	S	0.26 REF		
Mold Cap Height	M	0.53 REF		
Overall Length	D	11.00 BSC		
Overall Terminal Pitch	D1	9.75 BSC		
Overall Width	E	11.00 BSC		
Overall Terminal Pitch	E1	9.75 BSC		
Terminal Diameter	b	0.32	-	0.42

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only, displayed in parentheses.

196-Lead Thin Fine Pitch Ball Grid Array (4GB) - 11x11x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1		9.75	
Contact Pad Spacing	C2		9.75	
Contact Pad Width (X196)	X			0.45
Space Between Contact Pads	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23507 Rev B